# Branching Fidelity

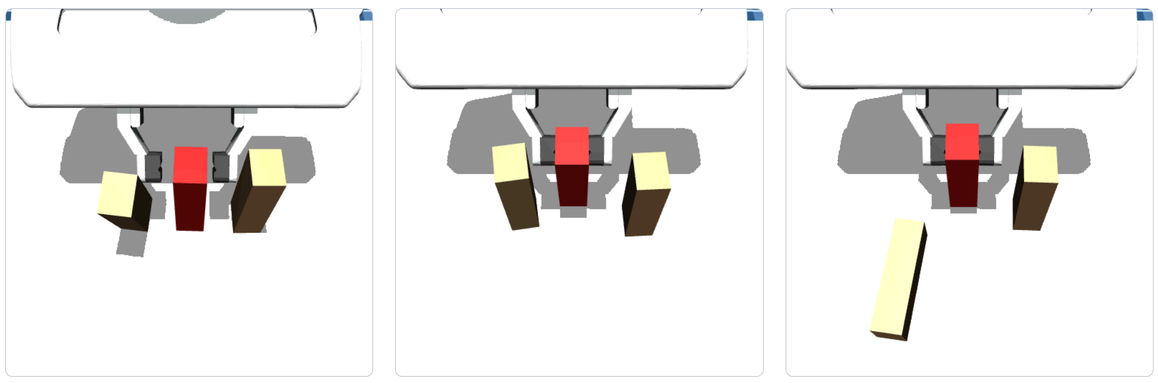
## Testing World Models as Intuitive Physics Engines for Robot Safety

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**A world model can predict well on average, even with the right amount of uncertainty, and still miss where physics branches. A safety check needs the second.**

## 1 The question

Before each action chunk, a robot should ask: **could my own small errors change what happens?** Picking the middle block, the same command lifts it cleanly or tips a neighbour, depending on a few millimetres of perception and execution error.

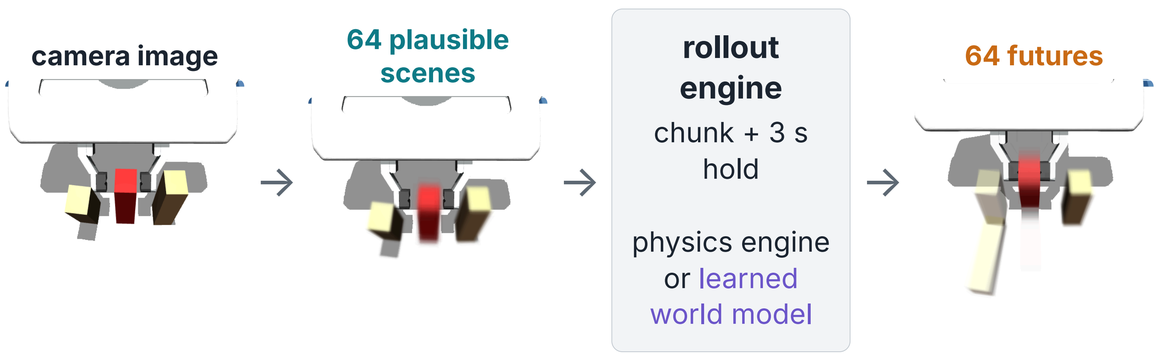


the chunk starts | 31 of 64 futures: neighbour stays | 33 of 64 futures: neighbour topples

The physics is deterministic: all uncertainty is the robot's own. A state is a **fork** if the outcome splits over futures sampled from that uncertainty.

## 2 The fork probe

Like the human intuitive physics engine: a few rough simulations over an uncertain scene; disagreement means precarious.




- Samples the robot's **measured** errors: perception 2.5 mm, execution about 3 mm.
- Flags the chunk if the 64 endings are **bimodal**, not merely different.
- No task labels, no failure data, no thresholds fit to labels.

## 3 With a physics engine, it works

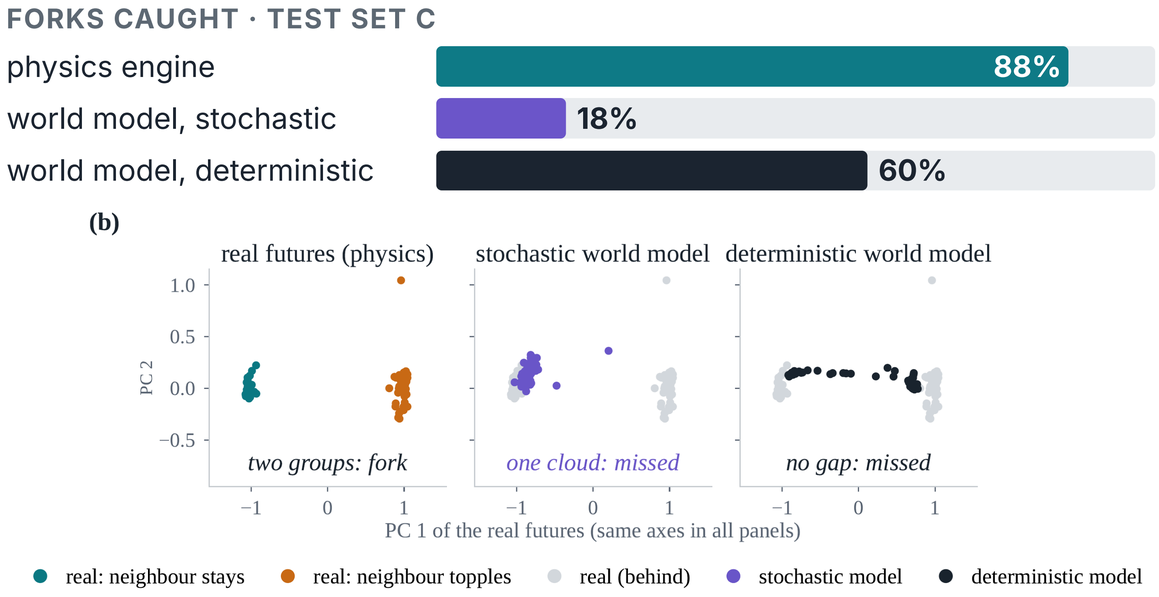
**88%** of forks caught with **7%** false alarms on unseen layouts, from one camera image (test set C, 156 states; 95% CI 77–94%)




Sampling perception error matters: it is as large as execution error. This is the bar a learned world model must reach.

## 4 Learned world models miss the branch

Replace the engine with a world model predicting in DINOv2 feature space. Both models below predict **3–4× better than a no-motion baseline**.




**Right amount, wrong shape.** The stochastic model's spread matches reality (0.98×), but only 15% lies along the branch (real: 57%), and it is barely larger on forks than on safe states (1.3× vs 4.8×). Five more generative variants catch at most 3%.

## 5 What closes the gap

**Uncertain starts, deterministic dynamics**, as in the intuitive physics engine: perturb the start, roll it out without noise. **Oversample rare branches**: find each training scene's rare branch with the same split test, no labels.




False alarms 7–11%. A generative model given the same uncertain starts still catches 0%: noise in the dynamics erases the forks.

## 6 Takeaways

**Measure branching fidelity**
Average error and calibrated spread do not show whether rollouts split where physics splits.

**Put uncertainty in the start**
Deterministic dynamics from perturbed starts keep the branch; per-step noise smears it.

**Train for rare branches**
Oversampling each scene's rare branch, found without labels, lifts catch rates.

**A fork is not a failure**
Pair the flag with a harm signal, such as whether the outcome can be undone, before acting.

Limitations: simulation, one three-block task; the ground truth uses the same error model the probe samples; single-seed, small world models.

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